

## EXPERIMENT 2

### EDA – DATA CLEANING

#### Aim

To clean the Customer Churn dataset by detecting and handling missing values, removing duplicate and unnecessary data, converting data types, and normalizing numerical data.

#### Procedure

1. Load the Customer Churn CSV dataset using Pandas.
2. Detect missing values using `isnull().sum()`.
3. Convert the TotalCharges column from object/string to numeric.
4. Fill missing TotalCharges values using the median.
5. Detect and remove duplicate records.
6. Remove the unnecessary customerID column.
7. Check the data types of all columns.
8. Apply Min-Max Scaling to selected numerical columns.
9. Apply Standardization using StandardScaler.
10. Display the cleaned and normalized data.

#### Code

```
import pandas as pd

from sklearn.preprocessing import MinMaxScaler, StandardScaler

# 1. Load dataset

df = pd.read_csv("WA_Fn-UseC_-Telco-Customer-Churn.csv")

print("Original Data:")

print(df.head())
```

```
# 2. Check missing values

print("\nMissing Values:")

print(df.isnull().sum())


# 3. Convert TotalCharges to numeric

df["TotalCharges"] = pd.to_numeric(
    df["TotalCharges"], errors="coerce"
)


# 4. Fill missing values

df["TotalCharges"] = df["TotalCharges"].fillna(
    df["TotalCharges"].median()
)


# 5. Check duplicates

print("\nDuplicate Rows:", df.duplicated().sum())


# 6. Remove duplicates

df = df.drop_duplicates()


# 7. Remove unnecessary column

df = df.drop("customerID", axis=1)


# 8. Check data types

print("\nData Types:")

print(df.dtypes)


# 9. Min-Max Scaling

scaler = MinMaxScaler()
```

```

df[["tenure", "MonthlyCharges", "TotalCharges"]] = scaler.fit_transform(
    df[["tenure", "MonthlyCharges", "TotalCharges"]]
)

print("\nAfter Min-Max Scaling:")
print(df[["tenure", "MonthlyCharges", "TotalCharges"]].head())

# 10. Standardization
standard_scaler = StandardScaler()

df[["tenure", "MonthlyCharges", "TotalCharges"]] =
standard_scaler.fit_transform(
    df[["tenure", "MonthlyCharges", "TotalCharges"]]
)

print("\nAfter Standardization:")
print(df[["tenure", "MonthlyCharges", "TotalCharges"]].head())

# 11. Final dataset information
print("\nFinal Dataset:")

df.info()

print("\nData Cleaning Completed!")

```

## OUTPUT

Original Data:

|              | customerID | gender | SeniorCitizen | Partner | Dependents | tenure |
|--------------|------------|--------|---------------|---------|------------|--------|
| PhoneService | \          |        |               |         |            |        |
| 0            | 7590-VHVEG | Female | 0             | Yes     | No         | 1      |
| No           |            |        |               |         |            |        |

|     |            |        |   |    |    |    |
|-----|------------|--------|---|----|----|----|
| 1   | 5575-GNVDE | Male   | 0 | No | No | 34 |
| Yes |            |        |   |    |    |    |
| 2   | 3668-QPYBK | Male   | 0 | No | No | 2  |
| Yes |            |        |   |    |    |    |
| 3   | 7795-CFOCW | Male   | 0 | No | No | 45 |
| No  |            |        |   |    |    |    |
| 4   | 9237-HQITU | Female | 0 | No | No | 2  |
| Yes |            |        |   |    |    |    |

| MultipleLines InternetService OnlineSecurity ... DeviceProtection |                  |             |     |     |     |     |
|-------------------------------------------------------------------|------------------|-------------|-----|-----|-----|-----|
| \                                                                 |                  |             |     |     |     |     |
| 0                                                                 | No phone service |             | DSL | No  | ... | No  |
| 1                                                                 | No               |             | DSL | Yes | ... | Yes |
| 2                                                                 | No               |             | DSL | Yes | ... | No  |
| 3                                                                 | No phone service |             | DSL | Yes | ... | Yes |
| 4                                                                 | No               | Fiber optic |     | No  | ... | No  |

| TechSupport StreamingTV StreamingMovies Contract |     |    |    |                |
|--------------------------------------------------|-----|----|----|----------------|
| PaperlessBilling \                               |     |    |    |                |
| 0                                                | No  | No | No | Month-to-month |
| Yes                                              |     |    |    |                |
| 1                                                | No  | No | No | One year       |
| No                                               |     |    |    |                |
| 2                                                | No  | No | No | Month-to-month |
| Yes                                              |     |    |    |                |
| 3                                                | Yes | No | No | One year       |
| No                                               |     |    |    |                |
| 4                                                | No  | No | No | Month-to-month |
| Yes                                              |     |    |    |                |

| PaymentMethod MonthlyCharges TotalCharges Churn |                           |       |         |     |
|-------------------------------------------------|---------------------------|-------|---------|-----|
| 0                                               | Electronic check          | 29.85 | 29.85   | No  |
| 1                                               | Mailed check              | 56.95 | 1889.5  | No  |
| 2                                               | Mailed check              | 53.85 | 108.15  | Yes |
| 3                                               | Bank transfer (automatic) | 42.30 | 1840.75 | No  |
| 4                                               | Electronic check          | 70.70 | 151.65  | Yes |

[5 rows x 21 columns]

Missing Values:

customerID 0

|                  |   |
|------------------|---|
| gender           | 0 |
| SeniorCitizen    | 0 |
| Partner          | 0 |
| Dependents       | 0 |
| tenure           | 0 |
| PhoneService     | 0 |
| MultipleLines    | 0 |
| InternetService  | 0 |
| OnlineSecurity   | 0 |
| OnlineBackup     | 0 |
| DeviceProtection | 0 |
| TechSupport      | 0 |
| StreamingTV      | 0 |
| StreamingMovies  | 0 |
| Contract         | 0 |
| PaperlessBilling | 0 |
| PaymentMethod    | 0 |
| MonthlyCharges   | 0 |
| TotalCharges     | 0 |
| Churn            | 0 |

dtype: int64

Duplicate Rows: 0

Data Types:

|                  |        |
|------------------|--------|
| gender           | object |
| SeniorCitizen    | int64  |
| Partner          | object |
| Dependents       | object |
| tenure           | int64  |
| PhoneService     | object |
| MultipleLines    | object |
| InternetService  | object |
| OnlineSecurity   | object |
| OnlineBackup     | object |
| DeviceProtection | object |
| TechSupport      | object |
| StreamingTV      | object |
| StreamingMovies  | object |
| Contract         | object |
| PaperlessBilling | object |

```

PaymentMethod      object
MonthlyCharges     float64
TotalCharges       float64
Churn              object
dtype: object

```

After Min-Max Scaling:

|   | tenure   | MonthlyCharges | TotalCharges |
|---|----------|----------------|--------------|
| 0 | 0.013889 | 0.115423       | 0.001275     |
| 1 | 0.472222 | 0.385075       | 0.215867     |
| 2 | 0.027778 | 0.354229       | 0.010310     |
| 3 | 0.625000 | 0.239303       | 0.210241     |
| 4 | 0.027778 | 0.521891       | 0.015330     |

After Standardization:

|   | tenure    | MonthlyCharges | TotalCharges |
|---|-----------|----------------|--------------|
| 0 | -1.277445 | -1.160323      | -0.994242    |
| 1 | 0.066327  | -0.259629      | -0.173244    |
| 2 | -1.236724 | -0.362660      | -0.959674    |
| 3 | 0.514251  | -0.746535      | -0.194766    |
| 4 | -1.236724 | 0.197365       | -0.940470    |

Final Dataset:

```
<class 'pandas.core.frame.DataFrame'>
```

RangeIndex: 7043 entries, 0 to 7042

Data columns (total 20 columns):

| #  | Column           | Non-Null Count | Dtype   |
|----|------------------|----------------|---------|
| 0  | gender           | 7043 non-null  | object  |
| 1  | SeniorCitizen    | 7043 non-null  | int64   |
| 2  | Partner          | 7043 non-null  | object  |
| 3  | Dependents       | 7043 non-null  | object  |
| 4  | tenure           | 7043 non-null  | float64 |
| 5  | PhoneService     | 7043 non-null  | object  |
| 6  | MultipleLines    | 7043 non-null  | object  |
| 7  | InternetService  | 7043 non-null  | object  |
| 8  | OnlineSecurity   | 7043 non-null  | object  |
| 9  | OnlineBackup     | 7043 non-null  | object  |
| 10 | DeviceProtection | 7043 non-null  | object  |
| 11 | TechSupport      | 7043 non-null  | object  |
| 12 | StreamingTV      | 7043 non-null  | object  |

```
13 StreamingMovies    7043 non-null    object
14 Contract           7043 non-null    object
15 PaperlessBilling   7043 non-null    object
16 PaymentMethod      7043 non-null    object
17 MonthlyCharges     7043 non-null    float64
18 TotalCharges       7043 non-null    float64
19 Churn              7043 non-null    object
dtypes: float64(3), int64(1), object(16)
memory usage: 1.1+ MB
None
```

Data Cleaning Completed!

### **Result**

The Customer Churn dataset was successfully cleaned by handling missing values, removing duplicates and unnecessary data, converting data types, and applying Min-Max Scaling and Standardization to numerical features.